

Question 1(a) Can this be done using the `sample` function? If yes, what table would we call `sample` on?
(1 point)

Yes, with "top_movies".

Question 1(b) Can this be done using the `np.random.choice` function? If yes, what array would we call `np.random.choice` on? **(1 point)**

Yes, with “studios_of_all_movies”.

Question 1(c) Can this be done using the `sample_proportions` function? If yes, what array would we call `sample_proportions` on? **(1 point)**

No.

Question 2(a) Can this be done using the `sample` function? If yes, what table would we call `sample` on?
(1 point)

Yes, with "top_movies".

Question 2(b) Can this be done using the `np.random.choice` function? If yes, what array would we call `np.random.choice` on? **(1 point)**

Yes, with “studios_of_all_movies”.

Question 2(c) Can this be done using the `sample_proportions` function? If yes, what array would we call `sample_proportions` on? **(1 point)**

Yes, with “studio_proportions_only”.

Question 5. Given your observed value, do you believe that Jade's model is reasonable, or is our alternative (that our deck is rigged) more likely? Explain your answer using the histogram of statistics simulated using Jade's model (produced above). **(4 Points)**

The histogram suggests that there is a higher likelihood of drawing a number card than a face card and that the probability of pulling certain cards is not equal. This contradicts Jade's model, which claims that it is equally likely to pull a number card or a face card. So, the alternative model, which claims that the deck is rigged and therefore not equal, is more reasonable.

